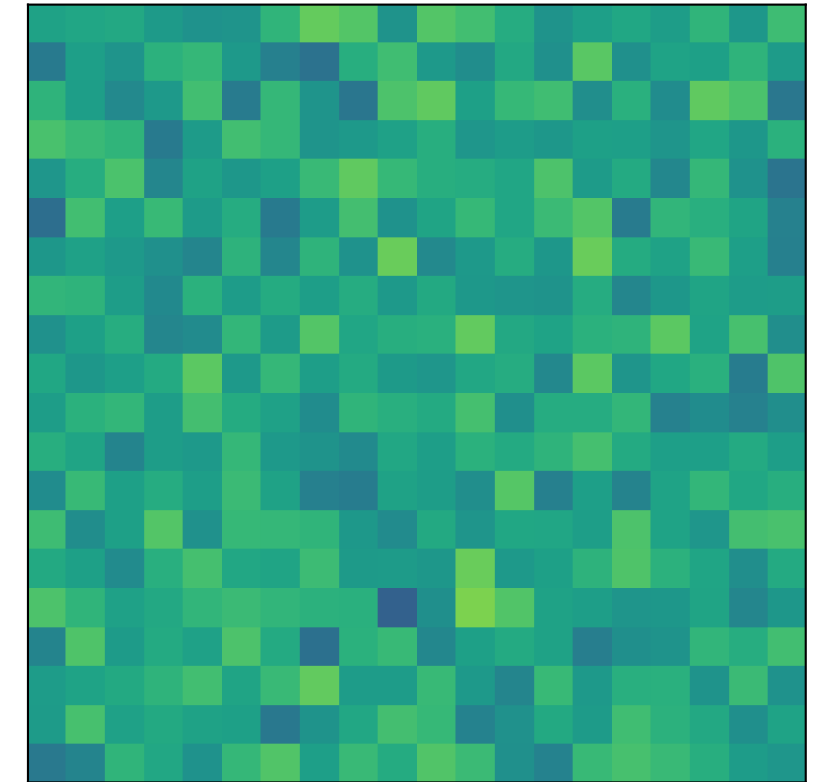
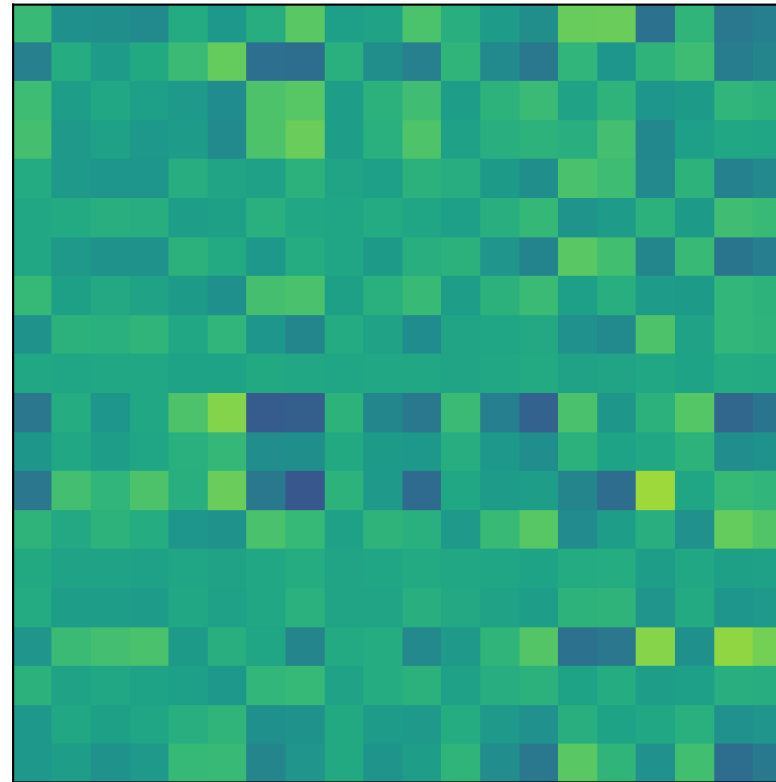
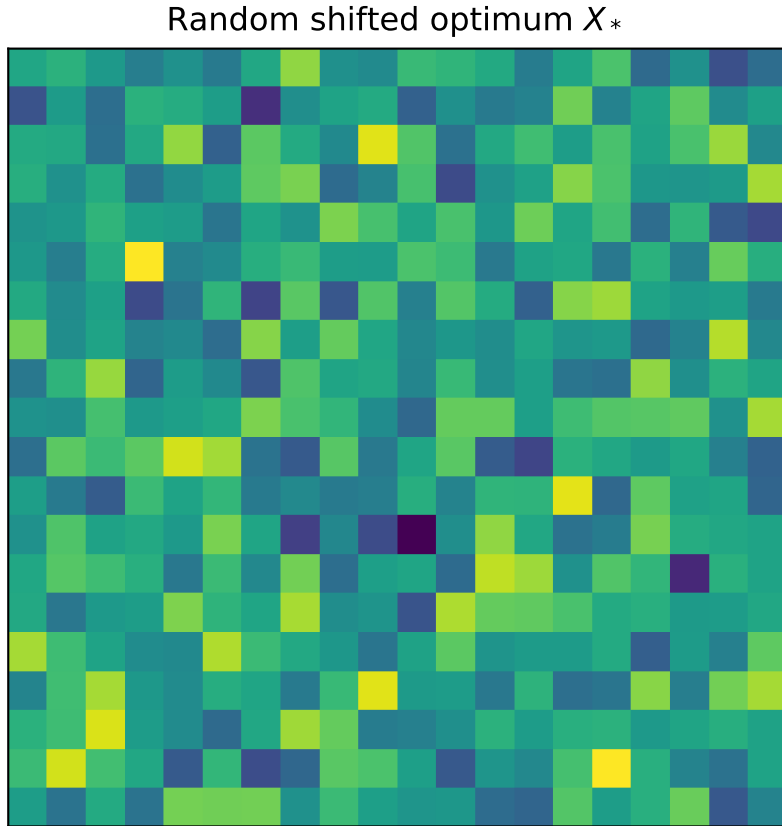


LoRA rank sweep on random full-rank shift
 $M = 20, D = M^2 = 400$, benchmark=ackley
 matched search dims: $r=1$: $d=40$, $r=2$: $d=80$, $r=4$: $d=160$
 LoRA SVD at $r=2, d = 2Mr = 80$ ($A \in \mathbb{R}^{20 \times 2}, B \in \mathbb{R}^{2 \times 20}$)
 rel. MSE=7.06e-01

Best RP at $r=2, d = 80$ ($P \in \mathbb{R}^{80 \times 400}$)
 rel. MSE=7.25e-01



Setup

Full space: $D = M^2 = 400$.

At each LoRA rank r , all methods search in the same dimension $d = 2Mr$:

LoRA: rank- r factors (A, B), $d = 2Mr$
 RP: random projection P , $d = 2Mr$
 RB: random blocking groups, $d = 2Mr$

Per-rank search dimensions:

$r=1 \rightarrow d=40$
 $r=2 \rightarrow d=80$
 $r=4 \rightarrow d=160$

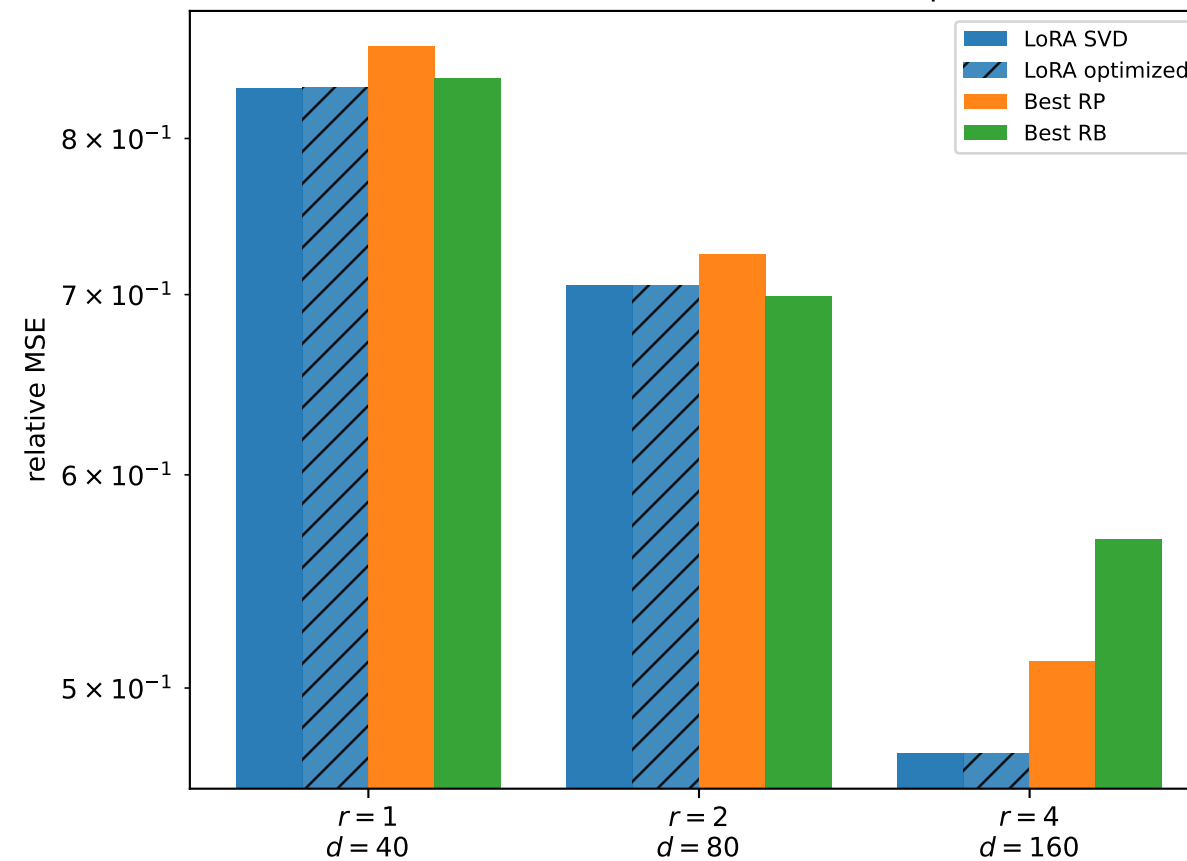
Random full-rank optimum.

Higher LoRA rank improves the best rank- r approximation (SVD lower bound).

RP/RB baselines are recomputed at each matched search dimension.

At $r=2$, RB uses $d = 80$ ($D = 400 \rightarrow \{0, \dots, 79\}$).

Reconstruction error vs rank (matched d per method)



Shifted ackley vs rank (matched d per method)

